NILAY TOSHNIWAL

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EDUCATION

Birla Institute of Technology & Science, Pilani (Goa Campus) *2023 – 2027*
B.E. (Hons.) Electronics & Communication Engineering, Finance Minor CGPA 7.18 / 10
Coursework: VLSI Design, Digital Design, Microprocessor & Interfacing, Analog Electronics, Signals & Systems, Digital Signal Processing, Machine Learning for Electronics, Probability & Statistics

Delhi Public School, Pune | Class XII (CBSE): 94.2% *2023*
Sarhad School | Class X (CBSE): 94.4% *2021*

PROJECTS

RISC-V System-on-Chip with AI and GPS Accelerators *Verilog, SoC Design*
- Designed a RISC-V processor in Verilog with two accelerators for jobs that are slow in software on a small chip: an 8-bit (INT8) neural-network engine and a GPS satellite-signal search, on a shared memory-mapped bus.
- **Ran the neural network 32.8x and the GPS search 5.6x faster than in software**, added low-power clock gating, and matched a C++ reference model on every instruction across 400 random test programs.

SlackSmith: AI-Proposed Chip Timing Fixes, Formally Checked *EDA, Formal Verification, team of 2*
- Built a loop where an AI model proposes fixes that make a chip design faster, and each fix must pass a formal proof (SymbiYosys) that the circuit's behaviour is unchanged, since wrong AI fixes look just like right ones.
- **The proof rejected 3 of 12 AI fixes,** one of which had passed 20,000 random tests; each kept fix is then re-timed.

Voice-Assistant Intent Routing in 6 Languages *NLP, LoRA Fine-Tuning*
- Fine-tuned a multilingual model (LoRA) to route voice-assistant requests to 54 intents in English and 5 Indian languages.
- **Recovered 83% of the accuracy lost to English-only training** by training on all six languages, and showed that a threshold for handing requests to a human, tuned in English, fails to carry over to Indian languages.

AXI4-Lite Clock-Domain-Crossing Bridge *SystemVerilog, UVM*
- Designed an AXI4-Lite bridge that safely passes data between parts of a chip on different clocks, via a Gray-coded FIFO.
- **Reached 100% functional coverage with zero failures** in UVM and SVA on Cadence Xcelium, after fixing 3 design bugs.

Amplifier and Voltage Regulator on a 130 nm Process *Analog Design, SKY130*
- Designed a two-stage amplifier (OTA) and a voltage regulator (LDO), sizing each transistor from measured gm/Id data.
- **Met the 60 dB gain target at all 15 process-temperature corners**; simulated offset matched a hand calculation to 4%.

EXPERIENCE

ElasticRun, *Research & Development Engineering Intern* *Jun 2026 – Aug 2026*
- **Built a two-armed warehouse robot in simulation** (ROS 2, MuJoCo) with inverse-kinematics arm control and a vision layer that flags damaged parcels at 95% accuracy and reads labels and barcodes.
- **Found 7 engine bugs** in the sorting simulator used in a vendor evaluation, and verified each fix with 1,341 regression tests.

MyJobFactory.com, *Backend Developer* *May 2025 – Jul 2025*
- Led a 6-person sub-team building a Sentence-BERT engine that ranks candidates for 500+ roles, **cutting screening time 40%.**

HONOURS & LEADERSHIP

Winner, TATA Vayu AI Studio Hackathon, Tata Communications national final *Jul 2026*
- **Led a team of 4 to build PCB Drishti Pro,** a YOLOv8 vision system that spots 6 kinds of manufacturing defect on bare circuit boards from a photo with no reference board needed, even on board designs it never saw in training.
- Put a rupee cost on every release, rework or scrap call, and served the model live on Tata Communications cloud.

Placed Second, APPCAIR JAW 2026 Hackathon, BITS Pilani *Aug 2026*
- **Scored 100/100 on the official test** by rebuilding a 41-table database from 687 raw documents and querying it.

Society of Automotive Engineers (SAE), BITS Pilani, *Core Member, Hardware & Sponsorship* *Aug 2024 – Aug 2025*
- Led hardware integration for 2 vehicle builds, fixing 12 critical electrical faults, and **raised ₹25L+ in sponsorship.**

Department of Finance & Marketing, BITS Goa, *Finance Head* *Aug 2024 – May 2026*
- **Allocated a ₹4-5 Cr budget** across 3 flagship fests with a 40-50 member core team, arbitrating claims between departments.

SKILLS

Hardware: Verilog, SystemVerilog, UVM, SVA, RISC-V, AXI4-Lite, clock-domain crossing, clock gating, synthesis (Yosys), static timing (OpenSTA), formal equivalence (SymbiYosys), Cadence Xcelium, Verilator, ngspice
AI / ML: PyTorch, Hugging Face Transformers, LoRA, INT8 quantization, YOLOv8, scikit-learn, RAG, LLM agents
Programming: Python, C, C++, Bash, SQL, Docker, FastAPI, MLflow, ROS 2, Git, Linux